Test Layer 3 Cross-Connect Traffic: test_l3.py

2025-09-15-22-07-01

Objective

The Layer 3 Traffic Generation Test is designed to test the performance of the Access Point by running layer 3 Cross-Connect Traffic. Layer-3 Cross-Connects represent a stream of data flowing through the system under test. A Cross-Connect (CX) is composed of two Endpoints, each of which is associated with a particular Port (physical or virtual interface).

Device Under Test Information

Device Under Test	DUT Name	
	DUT Hardware Version	
	DUT Software Version	
	DUT Serial Number	

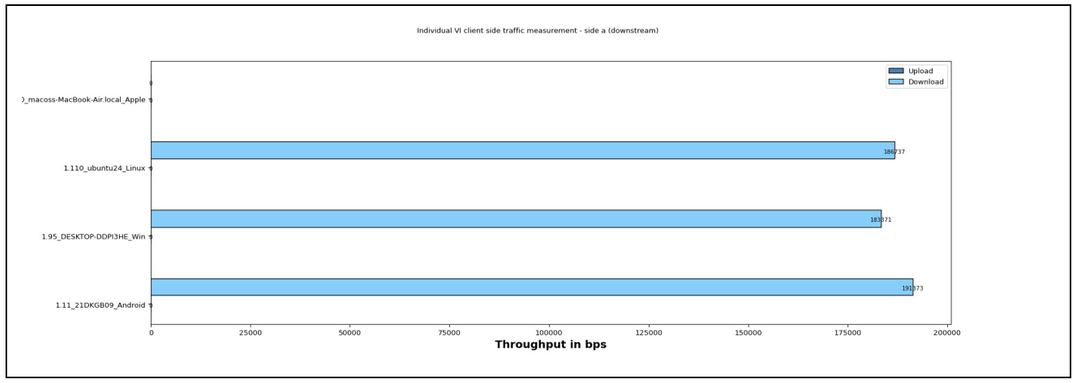
Test Configuration

Test Configuration	LANforge ip	192.168.204.75
	LANforge port	8080
	Upstream	1.1.eth1
	Test Duration	1m
	Polling Interval	5s
	Total No. of Devices	4

Radio Configuration

Individual throughput mcast download bps: 30000000 traffic VI (WiFi).

The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

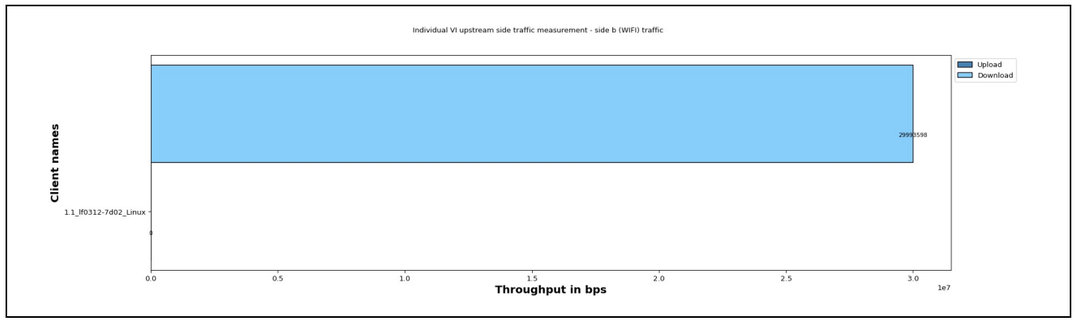


Client Alias	Host eid	Host Name	Device Type / Hw Ver	Endp Name	Port Name	Mode	Mac	SSID	Channel	Type of traffic	Traffic Protocol	Offered Upload Rate Per Client	Offered Download Rate Per Client	Upload Rate Per Client	Download Rate Per Client	Drop Percentage (%)
1.11_21DKGB09_Android	1.11	21DKGB09	samsung SM-M045F r14 sdk: 34	MLT-mrx-VI-wlan0-1	1.11.wlan0	802.11abgn 20	06:e8:6f:9c:30:7f	NETGEAR_2G_Open	11	VI	Mcast	0	30000000	0	191373	0.0
1.95_DESKTOP-	1.95	DESKTOP-	Win/x86	MLT-mrx-VI-	1.95.wlan0	802.11abgn	e4:a4:71:93:70:e7	NETGEAR_2G_wpa2	11	VI	Mcast	0	30000000	0	183371	0.0

DDPI3HE_Win		DDPI3HE	6.2	wlan0-2		20 1x1										
1.110_ubuntu24_Linux	1.110	ubuntu24	Linux/x86-64	MLT-mrx-VI-wlan0-3	1.110.wlan0	802.11bgn 20 2x2	5c:5f:67:9d:a3:86	NETGEAR_2G_wpa2	11	VI	Mcast	0	30000000	0	186737	0.0
1.400_macoss-MacBook-Air.local_Apple	1.400	macoss-MacBook-Air.local	Apple/x86-64	MLT-mrx-VI-en0-4	1.400.en0	802.11abgn-BE 20 2x2	30:35:ad:c5:e1:be	NETGEAR_2G_wpa2	11	VI	Mcast	0	30000000	0	0	0.0

Individual throughput upstream endp, offered upload bps: 0 offered download bps: 30000000 /station for traffic VI (WiFi).

The below graph represents individual throughput for 1 clients running VI (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



Client Alias	Host eid	Host Name	Device Type / HW Ver	Endp Name	Port Name	Mode	Mac	SSID	Channel	Type of traffic	Traffic Protocol	Offered Upload Rate Per Client	Offered Download Rate Per Client	Upload Rate Per Client	Download Rate Per Client	Drop Percentage (%)
1.1_lf0312-7d02_Linux	1.1	lf0312-7d02	Linux/x86-64	MLT-mtx-VI-eth1-0	1.1.eth1		00:03:2d:3f:7d:03		[channel: 1]	VI	Mcast	0	0	0	29993598	0.0

Layer 3 Cx Traffic Last Reporting Interval 1.1.eth1

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1757954291	09_15_2025_22_08_11	4	0	0	30000000	30000000	MTU	MTU	MTU	MTU	-1	1.1.eth1	58217	16845803	1 Gbps	1 Gbps	NaN	NaN	NaN	00:03:2d:3f:7d:03	[channel: 1]	0	0	0	0	0	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.11.wlan0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1757954291	09_15_2025_22_08_11	4	0	0	30000000	30000000	MTU	MTU	MTU	MTU	-1	1.11.wlan0	189700	33854	78 Mbps	78 Mbps	-31	94:A6:7E:74:26:21	802.11abgn 20	06:e8:6f:9c:30:7f	11	-80	0	191373	198770	992	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.95.wlan0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1757954291	09_15_2025_22_08_11	4	0	0	30000000	30000000	MTU	MTU	MTU	MTU	-1	1.95.wlan0	405129209	431021987	144.4 Mbps	144 Mbps	-13 dBm	94:A6:7E:74:26:22	802.11abgn 20 1x1	e4:a4:71:93:70:e7	11	105	2	183371	192532	953	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.110.wlan0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent

1757954291	09_15_2025_22_08_11	4	0	0	30000000	30000000	MTU	MTU	MTU	MTU	-1	1.110.wlan0	141359	60	144.4 Mbps	0 Mbps	-11 dBm	94:A6:7E:74:26:22	802.11bgn 20 2x2	5c:5f:67:9d:a3:86	11	168	2	186737	197271	971	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.400.en0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1757954291	09_15_2025_22_08_11	4	0	0	30000000	30000000	MTU	MTU	MTU	MTU	-1	1.400.en0	86016	1206	304.2 Mbps	145 Mbps	-26 dBm	94:A6:7E:74:26:22	802.11abgn-BE 20 2x2	30:35:ad:c5:e1:be	11	0	0	0	0	0	0.0	0	0	0	0